CSA0412-OPERATING SYSTEMS

EXP 11-Implement read and write using semaphore

```
#include <stdio.h> #include <pthread.h> #include <semaphore.h> #include <unistd.h>

sem_t mutex, wrt; int read_count = 0; int shared_data = 0;

void *reader(void *arg) { int id = *(int *)arg;

sem_wait(&mutex);
read_count++;

if (read_count == 1)
   sem_wait(&wrt);

sem_post(&mutex);

printf("Reader %d is reading data: %d\n", id, shared_data);
sleep(1);

sem_wait(&mutex);
read_count--;

if (read_count == 0)
   sem_post(&wrt);

sem_post(&mutex);

return NULL;


}

void *writer(void *arg) { int id = *(int *)arg;

sem_wait(&wrt);

shared_data++;
printf("Writer %d is writing data: %d\n", id, shared_data);
sleep(1);

sem_post(&wrt);
```

```
return NULL;


}

int main() { pthread_t r[3], w[2]; int rid[3] = {1, 2, 3}; int wid[2] = {1, 2}; int i;

sem_init(&mutex, 0, 1);
sem_init(&wrt, 0, 1);

for (i = 0; i < 3; i++)
   pthread_create(&r[i], NULL, reader, &rid[i]);

for (i = 0; i < 2; i++)
   pthread_create(&w[i], NULL, writer, &wid[i]);

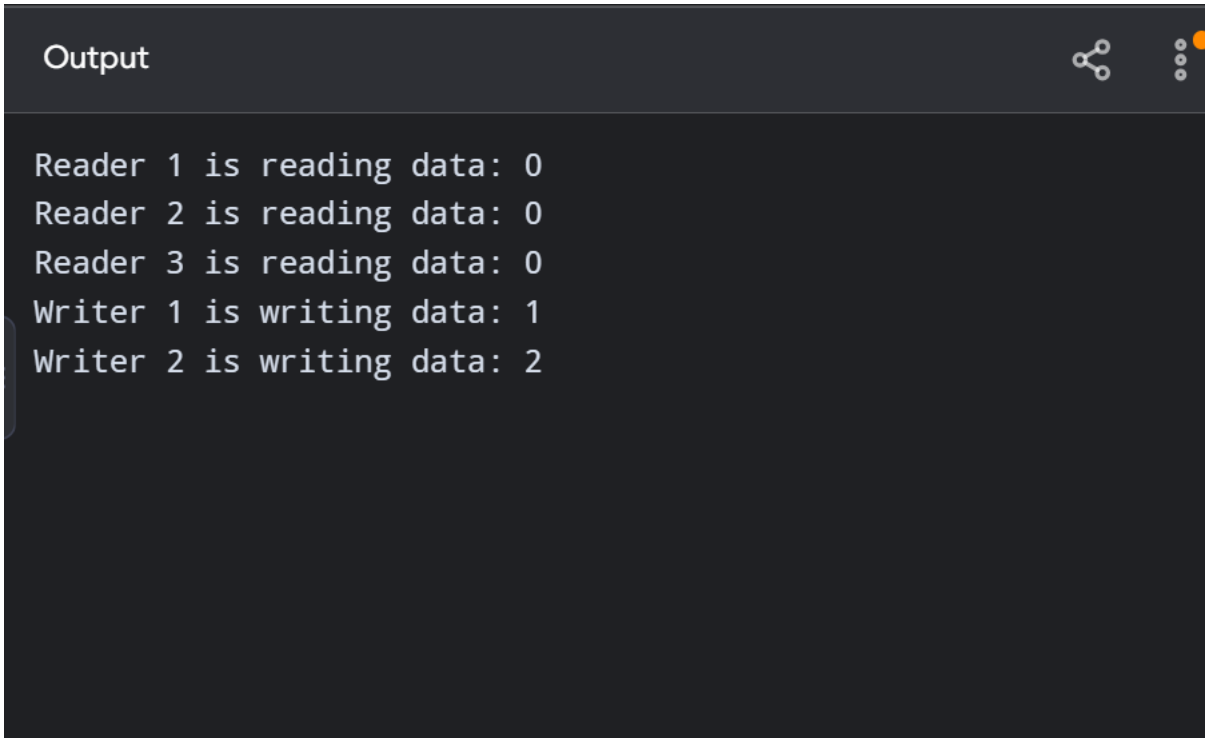
for (i = 0; i < 3; i++)
   pthread_join(r[i], NULL);

for (i = 0; i < 2; i++)
   pthread_join(w[i], NULL);

sem_destroy(&mutex);
sem_destroy(&wrt);

return 0;


}
```


Output
Reader 1 is reading data: 0
Reader 2 is reading data: 0
Reader 3 is reading data: 0
Writer 1 is writing data: 1
Writer 2 is writing data: 2